

access the Internet, you are presented with the option of allowing or blocking it. Even for configuring other parameters such as ports or protocols to block, you normally have a tabular or drop-down menu to select the options.

If, however, you need an example of how a great software can be killed by bad configuration parameters, don't look beyond eSafe. Getting this software to work was a big difficulty. The software gave an 'illegal operation' error on two machines due to problems with the Winsock implementation. Finally, we had to format our Test Bench again and load eSafe before other applications to get it working. Yes! All set, we thought... but no, that was just the beginning of the problems.

Try installing any software, and eSafe



## Supressing alerts in ZoneAlarm Pro

comes up with a number of messages that are plain irritating. You cannot stop the eSafe engine without deselecting the option in the Windows system configuration, which means that you have to restart the machine. Again, eSafe employs the good concept of user rights, wherein you can have different firewall settings for

different users based on the Windows login. But every time you change some configuration for the firewall, you have to again assign the configuration to the particular user for the changes to take effect.

Strangely enough, quite a few firewalls do not give the configuration option in the main toolbar itself. Instead, you have to navigate through some menus. In the case of Norton, you have to go through two levels, whereas Tiny has two separate windows for 'status' and 'administration'. ZoneAlarm and ZoneAlarm Pro are quite good in that respect with icons for different options arranged in the main window itself.

**Danger signs:** Personal firewalls do their work in the background while you are involved with some other application. The icon in the taskbar is probably the only way you come to know that the firewall is still alive. But then, how do you know when you are being attacked?

One way is to have a pop-up menu for every intrusion detected. But imagine being disturbed every 30 seconds with an alert screen just because somebody happened to access your shared folder in the local network. Alternatively, these firewalls indicate through flashing signs in the taskbar what kind of attack you are under.

The icons that Sygate used were among the most intuitive and were clearly visible even in the taskbar. Down and up arrows were used to denote inbound and outbound traffic, respectively, and red signals denoted blocked traffic or intrusion detection. Even with Norton, you could make out an alert from the way the icon flashed. ZoneAlarm had these bright yellow colour flashes that were hard to miss, although they didn't tell you about the nature of the attack. Some firewalls also allowed you to send an alert to an e-mail address or play an audible sound when an intrusion

[illegible]

Tiny is one of the free personal firewalls

occurred. But if you keep a high security level, be prepared for friends or colleagues putting up a No-honking Zone sign near you in protest.

**Keeping track:** Event logs are the best way to keep track of the activity in your disk. Basically, alerts are created on the basis of the rules you have created, and most firewalls allow you to log these alerts to a file. While McAfee has an activity log feature in its menu, the log file was one of the worst, filled with hex-dump and http headers of Internet activity that are of no use. Sygate kept the best logs. It had event logs in four categories: Security logs (notification of all intrusions), System logs (basic system information relating to the starting of Sygate), Traffic logs (details of the network and Internet traffic) and Packet logs (information on the packets transmitted, along with the hex-dump). All these details were given in neatly tabulated format, along with details of the local and remote host and the ports on which the connection was made. Just don't keep the file size for the packet logs too high, since it puts quite a bit of load on the system.

Even Norton had many logging options, but you can't exactly make out the fine distinctions between the different logs. BlackICE and Norton also provided a graphical representation of the intrusion attempts in terms of the history of the

## Reading the Warning Signals

| Firewall                      | Configuration ease | Intuitive Taskbar icons | Sounding alerts | Log file format | Filtering logs | Graphical representation |
|-------------------------------|--------------------|-------------------------|-----------------|-----------------|----------------|--------------------------|
| BlackICE Defender             | Partial            | ✗                       | ✓               | Average         | ✗              | ✓                        |
| eSafe Desktop                 | ✗                  | ✗                       | ✓               | Average         | ✗              | ✗                        |
| McAfee Personal Firewall      | ✗                  | ✗                       | ✗               | Poor            | ✗              | ✗                        |
| NeoWatch                      | ✓                  | ✗                       | ✓               | Good            | ✓              | ✗                        |
| Norton Personal Firewall 2001 | ✗                  | ✓                       | ✗               | Good            | ✓              | ✓                        |
| PGP Desktop Security          | ✗                  | ✗                       | ✓               | Average         | ✓              | ✗                        |
| Sygate Personal Firewall      | Partial            | ✓                       | ✓               | Good            | ✓              | ✓                        |
| Tiny Personal Firewall        | ✗                  | ✓                       | ✗               | Average         | ✗              | ✓                        |
| ZoneAlarm                     | ✓                  | ✓                       | ✓               | Average         | ✗              | ✓                        |
| ZoneAlarm Pro                 | ✓                  | ✓                       | ✓               | Average         | ✓              | ✓                        |